Khalid Karroum

Data Science & Machine Learning Master's Student

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PFE / Final-Year Internship Availability: January 2027 | 6 months

PROFESSIONAL SUMMARY

Master's student in Data Science and Information Systems Security with hands-on experience designing and deploying end-to-end machine learning and data analytics solutions. Skilled in Python, SQL, scikit-learn, PyTorch, data preprocessing, feature engineering, model evaluation, REST APIs, dashboards, Docker, and CI/CD. Built production-oriented projects in credit risk, customer churn, and network anomaly detection, combining predictive modeling with scalable backend and monitoring workflows. Seeking a 6-month PFE internship starting January 2027 in Data Science, Machine Learning, Data Analytics, or Data Engineering.

TECHNICAL SKILLS

Data Science & Machine Learning: Python, SQL, pandas, NumPy, scikit-learn, PyTorch, EDA, data cleaning, feature engineering, classification, regression, clustering, cross-validation, model calibration, model evaluation
Data & Engineering: PostgreSQL, MySQL, MongoDB, ETL/ELT, Data Warehousing, Django, FastAPI, REST APIs, Streamlit, Plotly
MLOps & Delivery: Docker, Docker Compose, GitHub Actions/CI, automated testing, drift monitoring, Git, GitHub
Security & Computer Science: Linux, Kali Linux, network security fundamentals, cryptography, data structures and algorithms, OOP, C, PHP

PROJECTS

Aegis-Credit - Credit Risk & Model Governance
Technologies: Python | scikit-learn | Django | PostgreSQL | REST API | Docker Compose | CI
- Built an end-to-end credit-risk screening and model-governance workflow with calibrated probabilities, human review cases, CSV/XLSX batch scoring, API-key REST scoring, PostgreSQL support, drift monitoring, Docker Compose, health checks, and CI.
- Compared Logistic Regression, Random Forest, and Gradient Boosting; the historical calibrated Gradient Boosting demo achieved 0.881 ROC-AUC on an untouched final test set. Documented data provenance and corrected feature-contract limitations requiring fresh release evaluation.

NetGuard - Network Anomaly Detection
Technologies: Python | scikit-learn | PyTorch | FastAPI | Streamlit | Plotly | Docker Compose
- Built an unsupervised network anomaly-detection workflow on UNSW-NB15 using the deployed Isolation Forest + LOF + PyTorch Autoencoder ensemble; labels are used only for evaluation.
- Served real-time single-connection scoring through FastAPI and built a Streamlit/Plotly operations dashboard; containerized backend and frontend services with Docker Compose.

Telco Churn Dashboard - Customer Churn Analytics
Technologies: Python | scikit-learn | Django | Streamlit | GitHub Actions/CI
- Built a reproducible churn pipeline with a stratified holdout, 5-fold cross-validation across Logistic Regression, Random Forest, and Gradient Boosting, sigmoid calibration, training-only threshold selection, and multi-metric evaluation.
- Reused one scoring and analytics layer across Django and Streamlit interfaces, with automated tests and GitHub Actions validating training and application behavior on every push and pull request.

EDUCATION

2025-2027 (Expected) | Master d'Excellence - Data Science & Information Systems Security
Sultan Moulay Slimane University, Beni Mellal, Morocco | Selected coursework: Advanced Machine Learning, Deep Learning, Reinforcement Learning, Advanced Data Analysis, Data Warehousing, Advanced Python/OOP

2024-2025 | Licence d'Excellence - Data Science & Information Systems Security
Sultan Moulay Slimane University, Beni Mellal, Morocco | Selected coursework: Machine Learning, Feature Engineering, Cryptography, Networks, Cybersecurity Fundamentals, Operational Research

2021-2024 | DEUG - Mathematics & Computer Science
Sultan Moulay Slimane University, Beni Mellal, Morocco

LANGUAGES

Arabic: Native | French: B1 (Intermediate) | English: B2 (Upper-Intermediate)